NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

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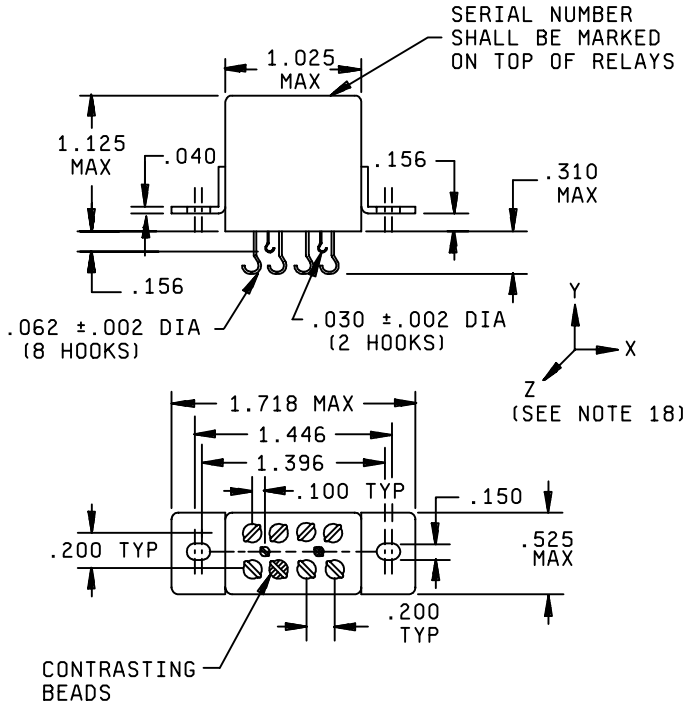
MIL-PRF-83536/13
27 March 1992

PERFORMANCE SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, ESTABLISHED RELIABILITY, DPDT, LOW LEVEL TO 10 AMPERES, MAGNETIC LATCH, HERMETICALLY SEALED, ALL WELDED, TRANSIENT SUPPRESSED DC COILS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation MIL-PRF-83536.

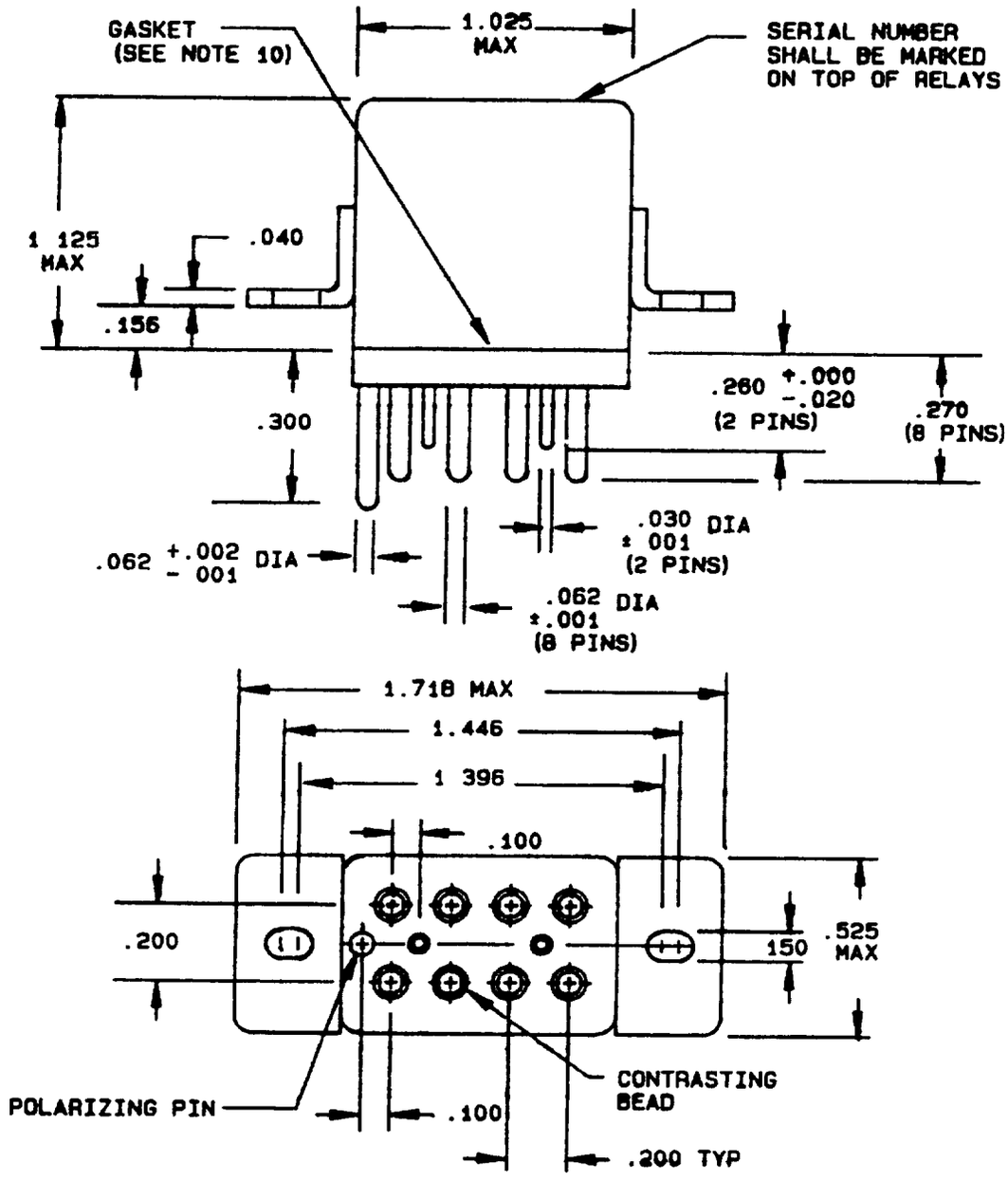


RAISED VERTICAL FLANGE MOUNT (RVFM) WITH SOLDER HOOKS

FIGURE 1. Dimensions and configurations.

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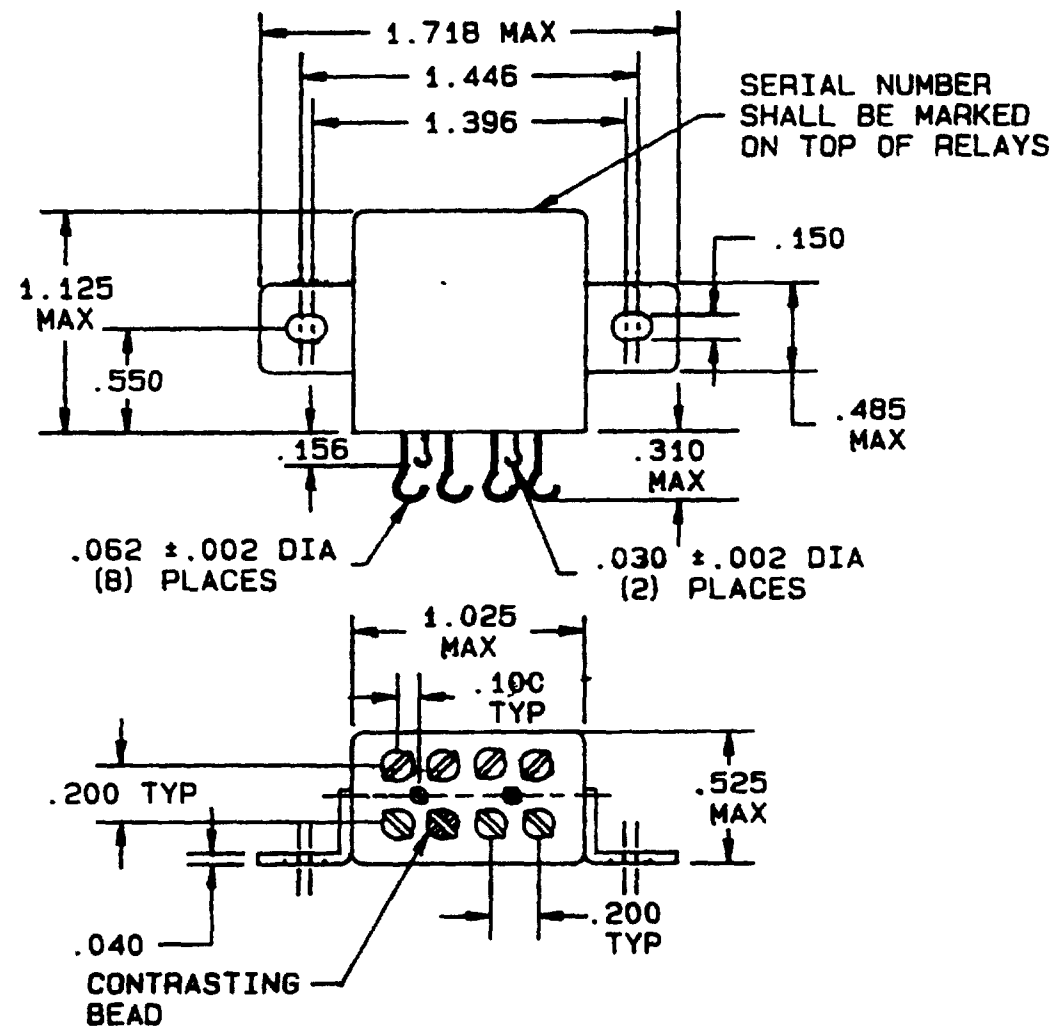
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RAISED VERTICAL FLANGE MOUNT (RVFM) WITH SOCKET PINS

FIGURE 1. Dimensions and configurations - Continued.

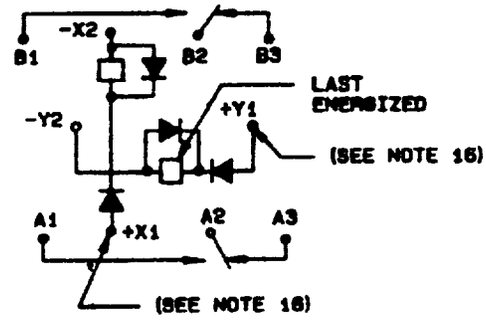
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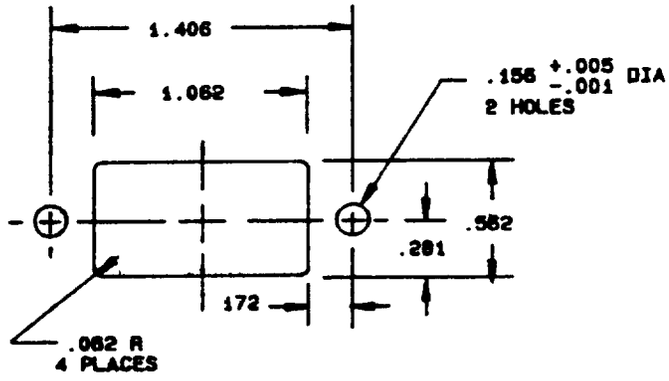
HORIZONTAL FLANGE MOUNT (HFM) WITH SOLDER HOOKS

FIGURE 1. Dimensions and configurations - Continued

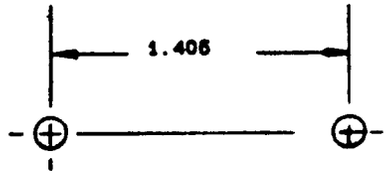
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CIRCUIT DIAGRAM (RVFM AND HFM)



MOUNTING HOLE LAYOUT (RVFM)



MOUNTING HOLE LAYOUT (HFM)

FIGURE 1. Dimensions and configurations - Continued.

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Inches	mm	Inches	mm	Inches	mm
.000	0.00	.156	3.96	.525	13.34
.001	0.03	.172	4.37	.562	14.27
.002	0.05	.200	5.08	1.025	26.04
.005	0.13	.260	6.60	1.062	26.98
.020	0.51	.270	6.86	1.125	28.58
.030	0.76	.281	7.14	1.396	35.46
.040	1.02	.300	7.62	1.406	35.71
.062	1.58	.310	7.87	1.446	36.73
.100	2.54	.485	12.32	1.718	43.64
.150	3.81	.500	12.70		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. There shall be affixed to the relay a suitable legible circuit diagram that identifies each terminal location specified.
5. These relays are polarized monostable.
6. This relay shall not be damaged by reverse polarity.
7. Applicable to horizontal flange mount only. The circuit diagram, manufacturer's PIN, and the military PIN shall be marked on the near side. The remaining portion of the nameplate data shall be marked on the far side.
8. Applicable to horizontal flange mount only Relays shall be marked with the manufacturer's name or CAGE code and date code. Marking shall be with the bottom of the print adjacent to the near side.
9. Socket pin terminals, except the polarizing pin, shall be gold-plated in accordance with MIL-G-45204, type II, class I. The nickel underplating shall be 50 to 150 microinches thick minimum.
10. Silicone rubber gasket, AMS 3332, shore hardness 15 to 35, thickness .050 ±.005 (1.27 ±0.13 mm).
11. Relay is magnetically latched in both positions.
12. All hooks shall be tin finished.
13. Relays shall have a (+) sign placed on the circuit diagram as shown.
14. Coil symbol optional in accordance with MIL-STD-1285.
15. Transient voltage (back EMF) 5 V dc maximum. Diodes shall have peak inverse voltage rating of 600 V dc.
16. Indicated terminal shall be identified by a contrasting bead.
17. Direction of hook openings is optional.
18. Plane of critical motion for vibration and shock is Y-axis.

FIGURE 1. Dimensions and configurations - Continued.

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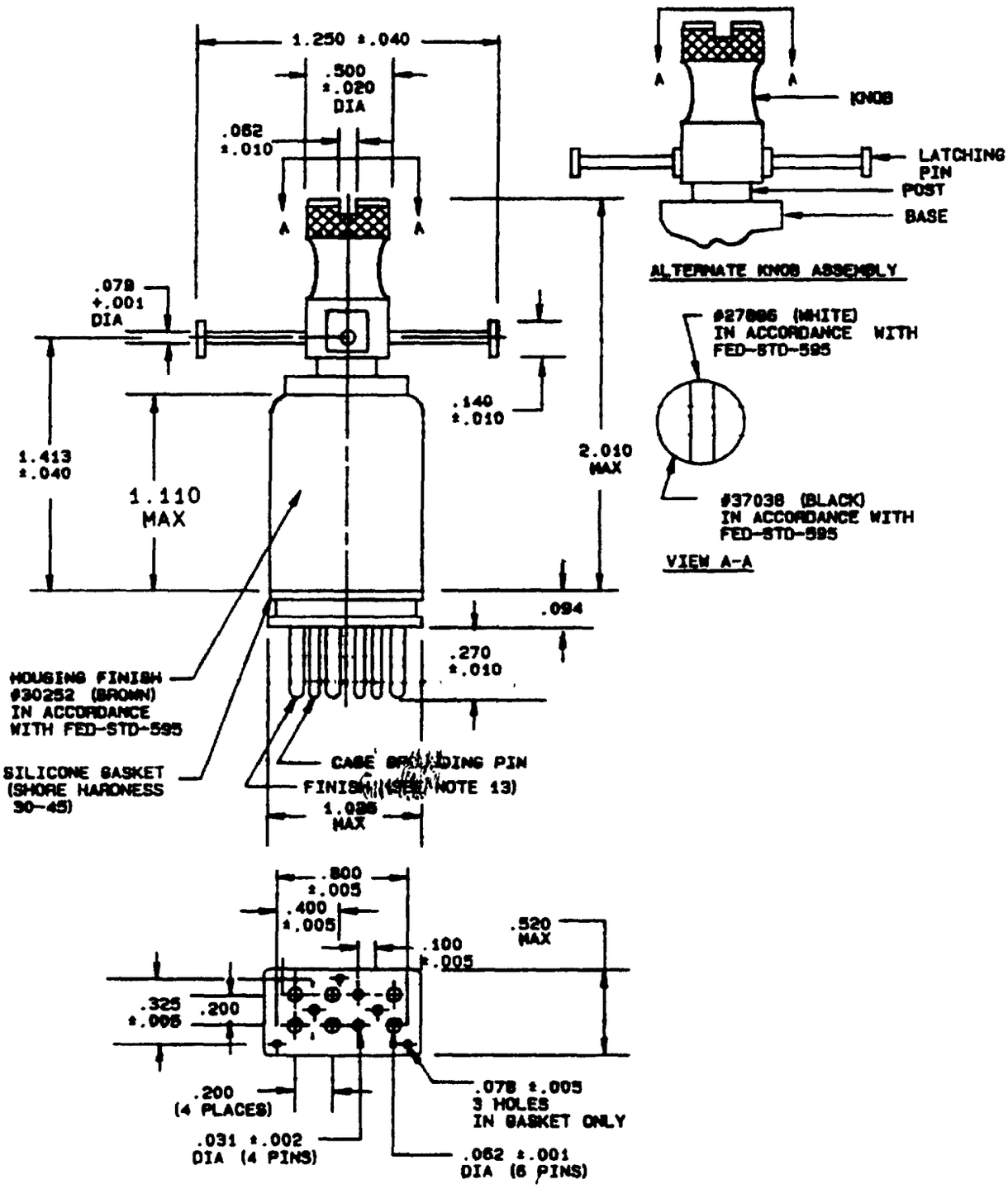
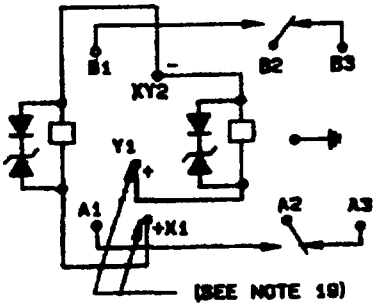


FIGURE 2. Outline dimensions and configuration, track mount (see notes 16 and 17).

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(SEE NOTES 4, 5, AND 6)

CIRCUIT DIAGRAM TRACK MOUNT

Inches	mm	Inches	mm
.001	0.03	.200	5.08
.005	0.13	.270	6.86
.010	0.25	.325	8.26
.020	0.51	.400	10.16
.031	0.79	.500	12.70
.040	1.02	.52	13.2
.062	1.57	.800	20.32
.078	1.98	1.025	26.04
.079	2.01	1.110	28.19
.094	2.39	1.250	31.75
.100	2.54	1.413	35.89
.140	3.56	2.010	51.05

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.
4. Relay is magnetically latched in both positions.
5. Caution note to observe polarity must appear on relays with dc coils.
6. Terminal numbers shall not appear on the header or on the gasket. There shall be affixed to the relay a suitable legible circuit diagram that identifies each terminal location specified.
7. Enclosure shall be brown 30252 in accordance with FED-STD-595.
8. With voltage applied to terminals X1-XY2 only, the relay contacts shall transfer and remain transferred regardless of whether the coil voltage is maintained beyond the operate time or removed.
9. With voltage applied to terminal Y1-XY2 only, the relay contacts shall transfer back to the condition existing before the voltage was applied to X1-XY2 and shall remain in this "re-transferred" condition regardless of whether the coil voltage is maintained beyond the operate time or removed.
10. With the same magnitude and polarity voltage applied to terminals X1-XY2 and Y1-XY2 simultaneously, the relay contacts shall not transfer from the condition existing prior to the application of voltage to both coils.
11. With 20 V dc maximum applied initially to terminals X1-XY2, the subsequent application of 28 V dc minimum to terminals Y1-XY2 (while the voltage is maintained on terminals X1-XY2) shall not cause the relay contacts to transfer from the condition resulting from the application of voltage to terminals X1-XY2 prior to the application of voltage to terminals Y1-XY2. Removal of voltage from terminals X1-XY2 (while voltage is maintained on terminals Y1-XY2) shall cause the relay contact transfer described in note 9.

FIGURE 2. Outline dimensions and configuration, track mount (see notes 16 and 17) - Continued.

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12. With 20 V dc maximum applied initially to terminals X1-XY2, the subsequent application of 28 V dc minimum to terminals X1-XY2 (while the voltage is maintained on terminals X1-XY2) shall not cause the relay contacts to transfer from the condition resulting from the application of voltage to terminals Y1-XY2 prior to the application of voltage to terminals X1-XY2. Removal of voltage from terminals Y1-XY2 (while the voltage is maintained on terminals X1-XY2) shall cause the relay contact transfer described in note 8.
13. All terminals shall be gold plated in accordance with MIL-G-45204, type II, class 1; underplating shall be nickel, 50 to 150 microinches thick.
14. To close A1 contact (operate) energize +X1 and +XY2. To close A1 and B1 contacts (operate/latch) energize X1/XY2 coil. To close A3 and B3 contacts (reset) energize Y1/XY2 coil.
15. Back EMF (transient voltage): 42 V dc maximum.
16 Track mount knob, post, and base shall be 303 Cres nickel plated in accordance with McAir P.S 13112 or equivalent, 100 microinches thick minimum. The latching pin shall be 300 Cres passivated or nickel plated in accordance with McAir P.S. 13112 or equivalent, 100 microinches thick minimum.
17. Track mount relay shall be capable of mating with M6106/4 "Bracket and Socket Assembly", configuration III. The latching pin shall be capable of withstanding an axial pull of 35 pounds, in a direction away from the knob, without degradation. The rotation of the knob mechanism shall be restricted such that the latching pins cannot be placed on the side opposite the M6106/4 bracket cutouts.
18. JANTX or equivalent screened semiconductors shall be used for internal coil suppression. Relays using suppression devices shall continue to operate should the suppression circuit be in failure mode. Diodes shall have a peak inverse voltage of 600 V dc minimum.
19. Indicated terminals shall be identified by contrasting beads.

FIGURE 2 Outline dimensions and configuration, track mount (see notes 16 and 17) - Continued.

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REQUIREMENTS:

Contact data:

Load ratings. See table I.

Low level: 10 to 50 µA at 10 to 50 mV dc or peak ac.

Intermediate current: Applicable.

Mixed loads: Applicable.

TABLE I. Rated contact load and cycles (amperes per pole).

Type of load (high level)	Cycles x 10^3	28 V dc	115 V ac 1 phase 400 Hz	115 V ac 1 phase 50/60 Hz 1/	115/200 V ac 3 phase 400 Hz	115/200 V ac 3 phase 50/60 Hz 1/
Resistive	100	30	30	2.5	30	2.5
Inductive	20	6	6	N/A	6	N/A
Inductive	10	N/A	N/A	2.5	N/A	2.5
Motor	100	4	4	2	4	2
Lamp	100	2	2	1	N/A	N/A

1/ For 50/60 Hz rating, rupture and overload are not applicable and life for each load shall be 10,000 cycles.

Life: 100,000 cycles, unless otherwise specified (see table I)

Contact voltage drop or resistance:

Initial contact voltage drop: 0.100 volt maximum Rated resistive current at 6 V dc or peak ac. Relays shall not make or break this load.

Initial contact resistance: 0.010 ohm maximum [illegible] at 6 V dc or peak ac

High level life (contact voltage drop):

During life: 10 percent open circuit voltage maximum

After life: 0.125 volt maximum.

Intermediate current (contact resistance):

During intermediate current: 3 ohms maximum

After intermediate current: 0.150 ohm maximum

Low level (contact resistance):

During low level: 100 ohms maximum.

After low level: 0.150 ohm maximum at 100 mA and 28 V dc

Contact bounce: 1.0 millisecond maximum.

Contact stabilization time: 2.5 milliseconds maximum.

Break bounce (normally open contacts): 0.1 millisecond maximum, (not applicable to track mount).

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Overload current: 40 amperes dc (60 amperes ac)

Rupture current: 50 amperes dc (80 amperes ac)

Time current relay characteristics: See table II.

TABLE II. Time current relay characteristics. 1/

Successive application	Amperes	Time
1	15	1 hour
2	50	5.0 seconds
3	100	1.2 seconds
4	250	0.2 second
5	350	0.1 second

1/ All relays shall withstand overload and fault currents. Relays must be able to sustain five applications (make and carry only) of power concurrently on adjacent poles at each of five different current levels in the sequence listed in table II. Separate relays shall be tested at 28 V dc and 115/200 V ac, 400 Hz 3-phase. Cooling time between successive applications shall be 30 minutes. Tests shall be performed on both normally open and normally closed contacts. There shall be no failures or evidence of welding or sticking and relays shall pass contact voltage drop at the conclusion.

Coil data 1/ 2/: See table III.

Operate time: 10 milliseconds maximum (each coil), 15 milliseconds maximum (each coil) (track mount) with rated coil voltage over the temperature range.

Duty rating: Continuous

Coil transient suppression: Applicable

Neutral screen: Applicable.

Electrical data:

Insulation resistance:

Initial: 100 megohms minimum at 500 V dc.

After life or environmental tests. 100 megohms minimum at 500 V dc (track mount), 50 megohms minimum at 500 V dc (all others).

1/ CAUTION: Due to possible interaction of relay magnetic fields, the following spacing requirements, as a minimum, shall be considered in dense packaging situations:

a. Row to row assisting fields. .125 (3.18 mm)

b. Row to row opposing fields: .1875 (4 763 mm)

c Side to side alternating fields: .0625 (1 588 mm)

d. Side to side like fields: .125 (3.18 mm).

2/ Track mount relays shall have the pickup voltage and timing tests performed with an M6106/4 bracket placed around the relay housing.

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TABLE III. Dash numbers and characteristics. 1/

Dash number 2/		Mount	Coil data						
			At 25°C				Over temperature range	Maximum pickup voltage	
Solder hook	Socket pin		Coil voltage (V dc) 3/		Coil resistance (ohms) minimum	Specified pickup voltage (V dc) 4/	Specified pickup voltage (V dc) 4/	High temperature test	Continuous current test
			Rated	Max					
001	002	RVFM	6	7.3	34	3.2	4.5	5.0	5.7
003	---	HFM	6	7.3	34	3.2	4.5	5.0	5.7
004	005	RVFM	12	14.5	135	6.5	9.0	9.9	11.25
006	---	HFM	12	14.5	135	6.5	9.0	9.9	11.25
007	008	RVFM	28	29 5/	400	14.5	18.0	19.8	22.5
009	---	HFM	28	29 5/	400	14.5	18.0	19.8	22.5
---	010	Track	28	29 5/	400	14.5	18.0	19.8	22.5
011	012	RVFM	48	50	1400	24.0	36.0	38.0	42.0
013	---	HFM	48	50	1400	24.0	36.0	38.0	42.0

1/ Each relay possesses high level and low level capabilities. However, relays previously tested or used above 10 mA resistive at 6 V dc maximum or peak ac open circuits are not recommended for subsequent use in low level applications.
2/ The suffix letter L, M, P, or R to designate the applicable failure rate level shall be added to the applicable listed dash number. Failure rate level (percent per 10,000 cycles): L, 3.0; M, 1.0; P, 0.1; R, 0.01. Example: 001L, 002R.
3/ CAUTION: The use of any coil voltages less than the rated coil voltage will compromise the operation of the relay.
4/ Pickup voltage as shown is for test purposes only and is not to be used for design criteria.
5/ When maximum ambient temperature does not exceed +85°C, the maximum coil voltage shall be 32 V dc.

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Dielectric withstanding voltage 3/:

	Track mount		All others	
	Coil to coil Coil to case	All other points	Coil to coil Coil to case	All other points
Sea level:				
Initial:	1050	1000	1000	1250
After life:	1250	1250	1000	1000
Altitude:				
At 80,000 feet:	350	350	250	350
At 300,000 feet:	500	500	500	500

Maximum leakage current: 100 microamperes rms in accordance with MIL-R-83536.

Environmental characteristics:

Temperature range: -70°C to +125°C.

Maximum altitude rating: 300,000 feet, 80,000 feet for track mount.

Shock (specified pulse): Applicable, MIL-STD-202, method 213, test condition C except peak value shall be 200 g's for 6 ±1 ms. Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts.

Vibration (sinusoidal): Applicable, MIL-STD-202, method 204, 30 g's except frequency range shall be 10 to 3,000 Hz. Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts.

Vibration (random): Applicable to track mount only. Applicable to qualification only. Test in accordance with MIL-STD-202, method 214, test condition IJ, 1.5 hours per axis. Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts.

Vibration (random): Applicable to qualification and group C inspection. Test in accordance with MIL-STD-202, method 214, test condition IG (0.4 g^2/Hz, 50 to 2000 Hz), 15 minutes each plane. Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts.

Acceleration: Applicable, except 15 g's.

Physical data:

Dimensions and configurations: See figure 1.

Weight: .090 pound (40.9 g's), 0.1 pound (45.5 g's) track mount.

Terminal strength:

Solder hook terminals:

Pull force:

For .030 diameter terminals: 3 pounds ±0.3 pound.

For .062 diameter terminals: 10 pounds ±1.0 pound.

3/ Dielectric may be improved by suitable insulation of terminals and wiring after installation.

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Bend: Applicable.

Twist: Applicable to coil terminals Y1 and Y2 only.

Socket pin terminals.

Pull force:

For .031 diameter terminals: 3 pounds ±0.3 pound (applicable to track mount).

For .062 diameter terminals: 10 pounds ±1.0 pound.

Bend: Applicable.

Twist: Applicable to coil terminals Y1 and Y2 only.

Terminal solderability: Applicable to solder hook terminals only.

Seal: Hermetic.

Marking: Applicable.

Part or Identifying Number (PIN): M83536/13 (dash number from table III and suffix letter designating failure rate level). PIN is a new term encompassing previous terms used in specifications such as part number, type designator, and identification number.

Qualification inspection:

Qualification inspection and sample size: See table IV.

TABLE IV. Qualification inspection and sample size. 1/

Single submission	Group submission	
18 units plus 1 open unit for level L at C = 0 2/	M83536/13-008	18 units plus 1 open unit for level L at C = 0 2/
33 units plus 1 open unit for level M at C = 0 2/	"	33 units plus 1 open unit for level M at C = 0 2/
Qualification inspection as applicable	"	Qualification inspection as applicable
	M83536/13-009 M83536/13-010	2 units, qualification inspection table, group II, shock, vibration, acceleration, terminal strength, and seal
	M83536/13-002 M83536/13-006 M83536/13-012	2 units, qualification inspection table, group II

1/ For retention of qualification or extension of qualification to lower failure rate levels, all life test data accumulated on MIL-R-83536/12 may be used in addition to MIL-R-83536/13 data. Prior to performance of retention of qualification testing, the relay manufacturer shall preselect the sampling plan.

2/ The number of units required for qualification testing shall be increased as required in group V, table II of MIL-R-83536, if the contractor elects to test the number of units permitting one or more failures. Prior to performance of qualification testing, the relay manufacturer shall preselect the sampling plan.

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Supersession data: See table V.

TABLE V. Supersession data.

Superseded PIN	Replacement PIN M83536/13-
M6106/50- 004	007
005	008
006	009
M6106/36- 001	010

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

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